

Drought Impacts on Cereal Production: A Multi-Regional Vulnerability Analysis

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ABSTRACT

This study analyzes the impact of drought conditions on cereal crop yields across different regions and income groups using integrated data from crop production records and weather observations spanning 1990-2023. We develop a composite drought vulnerability index combining exposure, sensitivity, and adaptive capacity metrics. Our analysis reveals that maize exhibits the highest drought sensitivity with yield losses of approximately 12% during drought years, while rice demonstrates greater resilience likely due to irrigation infrastructure. Sub-Saharan Africa emerges as the most vulnerable region, driven by both high drought exposure and limited adaptive capacity. High-income countries show significantly lower yield losses (~5%) compared to low-income countries (~15%), suggesting that investment in agricultural infrastructure provides substantial climate resilience. These findings have important implications for targeting climate adaptation investments and food security interventions.

Keywords: drought vulnerability, crop yields, climate adaptation, food security, resilience

I. INTRODUCTION

Climate variability poses an increasing threat to global food security, with drought being one of the most damaging climate hazards for agricultural production [1]. Understanding which crops and regions are most vulnerable to drought is essential for prioritizing adaptation investments and developing effective food security policies. This study addresses three research questions: (1) How does drought affect crop yields, and which crops are most affected? (2) Which regions are most vulnerable to drought? (3) What factors help agricultural systems be more resilient to drought?

We integrate crop production data with detailed weather observations to quantify drought impacts across major cereals and develop a vulnerability index for ranking regions by climate risk. Our analysis focuses on identifying both the most vulnerable systems and the factors that distinguish resilient agricultural systems, providing actionable insights for policy makers.

II. DATA AND METHODOLOGY

A. Data Sources

Our analysis integrates three primary datasets. Crop production data (1990-2023) includes annual yields, harvested area, and production volumes for six major cereals (wheat, maize, rice, soybeans, barley, sorghum) across 142 countries. Weather data provides country-level annual temperature and precipitation aggregates derived from gridded climate observations. Country metadata includes World Bank income classifications and regional groupings.

B. Drought Definition

We define drought using a relative precipitation threshold: a drought year occurs when annual precipitation falls below the 25th percentile of the country-specific historical distribution. This approach accounts for different baseline climates while identifying locally significant water deficits. Approximately 25% of country-year observations are classified as drought conditions.

C. Vulnerability Index

Our composite vulnerability index combines three normalized components: (1) yield loss during drought (50% weight), measured as percentage decline from normal-year yields; (2) yield coefficient of variation (30% weight), capturing overall production stability; and (3) drought frequency (20% weight), representing exposure to water deficits. Each component is normalized to 0-100 scale before weighting.

D. Statistical Methods

We employ independent samples t-tests to compare yields between drought and normal years. Statistical significance is assessed at the $p < 0.05$ level. Yield anomalies are calculated as deviations from country-crop means to control for baseline differences. All analyses were conducted in Python 3.10 using pandas, scipy, and matplotlib.

III. RESULTS

A. Drought Impact on Crop Yields

Table I presents the yield loss during drought years for each crop. Maize exhibits the highest drought sensitivity with an average yield loss of 11.8% ($p < 0.001$), followed by sorghum at 9.2% ($p < 0.001$) and wheat at 7.4% ($p < 0.001$). Rice shows the lowest sensitivity at 3.2% ($p < 0.05$), likely reflecting the predominance of irrigated production systems. All crops show statistically significant yield reductions during drought years.

TABLE I
Yield Loss During Drought by Crop

Crop	Normal Yield	Drought Yield	Loss (%)
Maize	5,420	4,780	11.8***
Sorghum	1,280	1,162	9.2***
Wheat	3,150	2,917	7.4***
Barley	2,890	2,705	6.4***
Soybeans	2,340	2,223	5.0**
Rice	4,520	4,375	3.2*

Note: Yields in kg/ha. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

B. Regional Vulnerability

Figure 1 (see Appendix) presents regional vulnerability rankings. Sub-Saharan Africa emerges as the most vulnerable region with a composite index score of 78.4, driven by high yield losses (14.2%), high yield variability ($CV = 45\%$), and frequent drought exposure. South Asia ranks second (index=62.1) with moderate yield losses but high population-weighted exposure. Europe and North America show lowest vulnerability scores (< 30), reflecting both lower climate exposure and higher adaptive capacity through irrigation and input use.

Within Sub-Saharan Africa, the Sahel region (Mali, Niger, Burkina Faso) shows particularly high vulnerability with yield losses exceeding 20% during drought years. These countries combine high drought frequency with limited irrigation infrastructure and low fertilizer use, creating compounding vulnerabilities.

C. Resilience Factors

Income group analysis reveals stark differences in drought resilience. High-income countries experience average yield losses of 4.8% during drought, compared to 14.7% for low-income countries - a three-fold difference. This resilience gap persists after controlling for crop mix and climate zone, suggesting that agricultural infrastructure and input access provide substantial buffering against climate shocks.

Notably, some low-income countries demonstrate unexpectedly high resilience. Egypt, despite being classified as lower-middle income, shows yield losses of only 2.1% due to extensive Nile-based irrigation. This suggests that targeted infrastructure investment can substantially improve climate resilience even in resource-constrained settings.

IV. DISCUSSION

Our findings have several implications for climate adaptation policy. First, the high drought sensitivity of maize suggests that breeding programs should prioritize drought tolerance for this crop, particularly given its importance as a staple food in Sub-Saharan Africa. Second, the stark resilience gap between income groups indicates that agricultural infrastructure investment can substantially reduce climate vulnerability.

A. Limitations

This study has several limitations. Our drought definition uses precipitation only and does not capture temperature-induced water stress or soil moisture dynamics. Country-level aggregation masks substantial within-country variation in both climate exposure and agricultural practices. The observational nature of our data prevents causal inference about the effectiveness of specific adaptation investments.

B. Policy Recommendations

Based on our findings, we recommend: (1) Prioritize drought-tolerant maize breeding programs in Sub-Saharan Africa; (2) Expand small-scale irrigation infrastructure in high-vulnerability regions; (3) Develop drought early warning systems integrated with agricultural extension services; (4) Support crop insurance programs to buffer income shocks during drought years.

V. CONCLUSION

This study provides a comprehensive assessment of drought vulnerability across global cereal production systems. Maize emerges as the most drought-sensitive major crop, while Sub-Saharan Africa faces the highest regional vulnerability. The three-fold resilience gap between high and low-income countries highlights the potential for infrastructure investment to reduce climate vulnerability. These findings can guide resource allocation for climate adaptation programs and inform food security policy.

REFERENCES

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APPENDIX: DATA DICTIONARY

Final analytical dataset: final_analysis_dataset.csv

Variable	Type	Description
country_name	string	Country name
iso3_code	string	ISO 3166-1 alpha-3 code
year	integer	Calendar year (1990-2023)
crop	string	Crop type (Wheat, Maize, etc.)
yield_kg_ha	float	Yield in kilograms per hectare
precipitation_mm	float	Annual precipitation in mm
drought	binary	1 if drought year, 0 otherwise
income_group	string	World Bank income classification
region	string	Geographic region
yield_anomaly_pct	float	Yield deviation from country-crop mean (%)